

Maurizio Ungaro

Scientific CV

Staff Scientist, [Jefferson Lab](#) | Nuclear physicist and simulation software developer
ungaro@jlab.org maureeungaro.github.io/home github.com/maureeungaro
Google Scholar INSPIRE ORCID 0000-0001-6982-3310 LinkedIn

Profile

Nuclear physicist and simulation software developer at Jefferson Lab working on Geant4-based detector simulations, [GEMC](#), CLAS12 production workflows, detector systems, and nucleon-structure research. My work connects physics analysis, detector operations, and simulation infrastructure, with a focus on making Geant4 workflows easier to build, run, document, and share.

Current Position

Staff Scientist

2011–present

[Jefferson Lab](#), Experimental Hall-B, Newport News, VA

Hall-B and CLAS12 detector operations, simulation software, Geant4 support, LTCC calibration and performance, distributed simulation production, and nucleon-structure analysis.

Previous Positions

Postdoctoral Research Associate and Research Associate

2004–2011

University of Connecticut, Storrs, CT

CLAS analyses, meson electro-production, detector software, and CLAS12 simulation and reconstruction development.

Adjunct Assistant Professor of Physics

2011

Christopher Newport University, Newport News, VA

Undergraduate physics laboratory instruction.

Education

2003 Ph.D. in Nuclear Physics, [Rensselaer Polytechnic Institute](#), Troy, NY. Thesis: π^0 electro-production from $\Delta(1232)$ at high momentum transfer.

1999 Laurea in Fisica, [Universita degli Studi di Genova](#), Genoa, Italy.

Research and Technical Leadership

- Nucleon-structure measurements using meson electro-production and CLAS/CLAS12 data.
- Development and support of [GEMC](#), a database-driven Geant4 simulation workflow.
- CLAS12 simulation releases, geometry tags, and production workflows through [CLAS12 Simulations](#).
- Distributed simulation production using HTCondor and Open Science Grid resources.
- CLAS12 [Low Threshold Cherenkov Counter](#) operation, maintenance, calibration, and detector-performance studies.
- [Geant4 at JLab](#) tutorials, examples, and user support for Jefferson Lab users.

Selected Projects

GEMC: Database-driven Geant4 simulations with a Python-friendly workflow.

<https://gemc.github.io/home/>

Geant4 at JLab: Tutorials, examples, and support material for Jefferson Lab users.

<https://jeffersonlab.github.io/g4home/>

CLAS12 Simulations: Tagged CLAS12 geometry and simulation releases.

<https://github.com/gemc/clas12Tags>

CLAS12 on OSG: Distributed CLAS12 simulation production workflows.

<https://maureeungaro.github.io/home/osg/osg>

Low Threshold Cherenkov Counter: CLAS12 forward detector work for pion/kaon discrimination.

<https://maureeungaro.github.io/home/ltcc/ltcc>

Selected Collaborations and Facilities

- Jefferson Lab and Jefferson Lab Hall-B: current laboratory and experimental program.
- Heavy Photon Search: dark-sector search collaboration at Jefferson Lab.
- SPring-8: prior experimental work in Japan through LEPS/SPring-8 collaboration.

Teaching and Mentoring Experience

- Adjunct Assistant Professor of Physics, Christopher Newport University, 2011.
- Instructor, PHYS 105 Elementary Physics Laboratory, Christopher Newport University, 2011.
- Instructor, PHYS 152 Intermediate Physics Laboratory, Christopher Newport University, 2011.
- Teaching Assistant, Physics I, Rensselaer Polytechnic Institute, Spring 2000.
- Teaching Assistant, Electromagnetic Theory, Rensselaer Polytechnic Institute, Fall 2000.
- Supervised undergraduate and graduate students on Jefferson Lab data analysis, detector calibration, and CLAS/CLAS12 physics analyses.
- Mentored student projects in meson electro-production, semi-inclusive reactions, target-spin asymmetries, quark propagation in nuclei, and detector simulation.

Selected Approved Proposals and Experimental Programs

- Hard exclusive electroproduction of π^0 and η with CLAS12, E12-06-108, spokesperson.
- Beyond the Born Approximation: precise comparison of e^+p and e^-p elastic scattering in CLAS, E07-005.
- Heavy Photon Search at Jefferson Laboratory, E12-11-006.
- Meson spectroscopy with low- Q^2 electron scattering in CLAS12, E12-11-005.
- Nucleon resonance studies with CLAS12, E12-09-003.
- Deeply Virtual Compton Scattering with CLAS at 11 GeV, E12-06-119.

Selected Plenary and Invited Talks

- Electromagnetic $N \rightarrow N^*$ excitations, Photonuclear Reactions Gordon Research Conference, Tilton, NH, August 2010.
- CLAS data and progress in the investigation of electro-excitation of nucleon resonances, 4th Workshop on Exclusive Reactions at High Momentum Transfer, Newport News, VA, May 2010.
- π^0 electro-production from $\Delta(1232)$ at high Q^2 , QCD 2008, Montpellier, France, July 2008.
- π^0 electro-production from $\Delta(1232)$ at high Q^2 , Baryons 2007, Seoul, Korea, June 2007.

- The two-photon exchange mechanism, Research Centre for Nuclear Physics, Osaka, Japan, November 2006.
- N^* at Jefferson Lab: exploring the high- Q^2 regime, Structure of Hadrons, Athens, Greece, June 2006.
- The $N \rightarrow \Delta$ transition, Jefferson Lab Workshop, Newport News, VA, June 2005.

Technical Skills

Simulation and analysis: Geant4, GEMC, CLAS12 simulations, ROOT, detector geometry, digitization workflows, PyVista visualization.

Programming and infrastructure: C++, Python, shell scripting, Git, GitHub, CI, Docker, Meson, CMake, SCons.

Scientific computing: HTCondor, Open Science Grid, SQLite, CSV/JSON/ROOT workflows, reproducible environments, documentation.

Communication: tutorials, technical writing, presentations, mentoring.

Selected List of Publications

- **M. Ungaro.** “Geant4 Monte-Carlo (GEMC): A database-driven simulation program.” *EPJ Web of Conferences* 295, 05005 (2024). doi:10.1051/epjconf/202429505005.
- **M. Ungaro et al..** “The CLAS12 Geant4 simulation.” *Nucl. Instrum. Meth. A* 959 (2020). doi:10.1016/j.nima.2020.163422.
- **M. Ungaro et al..** “The CLAS12 Low Threshold Cherenkov detector.” *Nucl. Instrum. Meth. A* 957 (2020). doi:10.1016/j.nima.2020.163420.
- **M. Ungaro, K. Joo et al..** “Measurement of the $N \rightarrow \Delta(1232)$ transition at high momentum transfer by π^0 electroproduction.” *Phys. Rev. Lett.* 97, 112003 (2006). doi:10.1103/PhysRevLett.97.112003.
- **I. G. Aznauryan, K. Joo, M. Ungaro et al..** “Electroexcitation of nucleon resonances from CLAS data on single pion electroproduction.” *Phys. Rev. C* 80, 055203 (2009). doi:10.1103/PhysRevC.80.055203.
- **F. X. Girod, K. Joo, M. Ungaro et al..** “Deeply virtual Compton scattering beam-spin asymmetries.” *Phys. Rev. Lett.* 100, 162002 (2008). doi:10.1103/PhysRevLett.100.162002.
- **K. Joo, M. Ungaro et al..** “Beam spin asymmetry measurements from deeply virtual meson production.” *AIP Conf. Proc.* 915, 607 (2007). doi:10.1063/1.2750854.
- **M. Ungaro, K. Joo and P. Stoler.** “ $\gamma^* N \rightarrow \Delta$ at JLab: Exploring the high- Q^2 regime.” *AIP Conf. Proc.* 904, 232 (2007). doi:10.1063/1.2734308.

Online Profiles

maureeungaro.github.io/home github.com/maureeungaro Google Scholar INSPIRE ORCID 0000-0001-6982-3310